Duality

A linear map $V \to F$ turns each vector into a single number. The i-th coordinate in a basis does this, and so do evaluating a polynomial at a point, integrating a function, and taking the trace of a matrix. These **linear functionals** are the measurements we can make on a vector space, and together they form a vector space of their own, the dual space V^*.

The dual space mirrors V with every inclusion reversed. A subspace corresponds to the functionals that vanish on it. An injective map corresponds to a surjective one. The kernel of a map corresponds to the image of its dual map, whose matrix is the transpose. Passing to the dual turns a hard question into an easy one surprisingly often. It also explains facts we already know from a new angle: why a subspace of F^n is always the solution set of a homogeneous system, and why row rank equals column rank.

What you need. From Chapter 0, the transpose and its arithmetic (Definition 0.9.10, Theorem 0.9.5) and matrix multiplication read column by column (Theorem 0.9.3), together with polynomials (Definition 0.8.1), which supply the infinite-dimensional example of §01. From Chapter 1, bases, dimension, and complements (Definition 1.5.1, Definition 1.6.1, Theorem 1.7.6), and bases of infinite-dimensional spaces (Theorem 1.8.2) for the remarks on infinite dimension. From Chapter 2, linear maps and the space $\mathcal{L}(V, W)$ (Definition 2.1.1, Theorem 2.4.1) and the Rank–Nullity Theorem (Theorem 2.3.1). From Chapter 3, the row space and null space of a matrix (Definition 3.5.2, Definition 3.5.3) and inverses by row reduction (Theorem 3.4.6). From Chapter 4, the matrix of a map and change of basis (Definition 4.1.1, Theorem 4.2.1), and quotient spaces with the First Isomorphism Theorem (Definition 4.4.3, Theorem 4.4.7).

5.-1.1 Roadmap

- **Linear functionals and the dual space.** Measurements as vectors; dual bases; separating vectors by functionals.
- **Annihilators.** The functionals that vanish on a subspace, and the formula $\dim U + \dim U^0 = \dim V$.
- **The double dual.** A natural isomorphism $V \to V^{**}$ that needs no basis, and what "natural" means.
- **The dual map and the transpose.** Pulling functionals back along a map; its matrix is the transpose; kernels and images trade places.
- **Duality for subspaces and quotients.** The dual of a subspace is a quotient of the dual, and the dual of a quotient is an annihilator; hyperplanes and codimension.
- **Hyperplanes and linear systems revisited.** Every subspace is cut out by equations; converting between parametric and implicit descriptions.

5.-1.2 Named moves

- **Test against functionals.** To show a vector is zero, show that every functional (or every vector of a dual basis) sends it to zero.
- **Pass to the dual and count.** Replace a subspace by its annihilator and use $\dim U + \dim U^0 = \dim V$.
- **Natural or chosen?** Ask whether an isomorphism needs a basis. $V \cong V^*$ does; $V \cong V^{**}$ does not.
- **Dualize a statement.** Injective becomes surjective, kernel becomes image, a matrix becomes its transpose, and inclusions reverse.